Extended Euclidean algorithm to find Greatest Common Divisor (GCD)

Module I

Extended Euclidean Algorithm

The extended Euclidean algorithm is an algorithm to compute integers x and y such that, **ax + by = gcd(a,b)**

- By reversing the steps in Euclidean algorithm, it is possible to find these integers x and y.
- The whole idea is to start with the GCD and recursively work way backwards.

Example

Example1: gcd (12, 34) ? gcd(a,b), a=12 and b=34.

$\frac{a}{n}, a = nq + r \; from \; division \; algorithm, \frac{34}{12}.$

$$34 = 2 * 12 + 10$$

$$12 = 10 * 1 + 2(gcd)$$

$$10 = 2 * 5 + 0$$

2= 12 – 1*10 → E.q.1

10= 34 – 2*12 → E.q.2

Substitute e.q 2 in e.q 1

2= 12 – 1*(34 – 2*12) → 2= 12 – (1*34) + 2*12) → 2= 3*12 – (1*34) → x=3 and y=-1

Example

Example2: gcd (93, 219) ? gcd(a,b) = 93x+219y, a=93 and b=219.

3 = 93x+219y

$\frac{a}{n}, a = nq + r \; from \; division \; algorithm, \frac{219}{93}.$

$$219 = 93 * 2 \ + 33 \rightarrow 93 = 33 \ * 2 + 27 \rightarrow 33 = 27 \ * 1 + 6$$

$$27 = 6 * 4 \ + 3(gcd) \rightarrow 6 = 3 * 2 \ + 0.$$

Reverse the operation,

3 = 27 – 4x6

= 27 – 4x(33-1x27)

= 27 – 4x33 + 4x27

= – 4x33 + 5x27

= – 4x33 + 5x(93-2*33)

= 5x93 – 14x33

= 5x93 – 14x(219-2*93)

= 5x93 – 14x219 + 28x93

3 = 33x93 – 14x219, x= 33 and y= -14